American International University Of Bangladesh (AIUB)

Department of Computer Science(CS)

Semester: Spring 2025-26

Course code: *00501* **Course:** *COMPUTER GRAPHICS*

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Section: *I*

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Submission Date: ___*/03/2026*

```
#include <windows.h> // for MS Windows

#include <GL/glut.h> // GLUT, include
glu.h and gl.h

void renderBitmapString(float x, float y,
float z, void *font, char *string) {

   char *c;

   glRasterPos3f(x, y,z);

   for (c=string; *c != '\0'; c++)

   {

      glutBitmapCharacter(font, *c);

   }

}

void display() {

   glClearColor(0.8f, 0.8f, 0.8f,0.0f);
glClear(GL_COLOR_BUFFER_BIT);
```

//Finland

```
glBegin(GL_LINES);

 glColor3f(1.0f,0.0f,0.0f);

 glVertex2f(0.0f,1.0f);

 glVertex2f(0.0f,-1.0f);

 glVertex2f(-1.0f,0.0f);

 glVertex2f(1.0f,0.0f);

glEnd();


glBegin(GL_QUADS);

glColor3f(0.15f,0.14f,0.76f);

glVertex2f(0.1f,0.1f);

glVertex2f(0.9f,0.1f);

glVertex2f(0.9f,0.7f);

glVertex2f(0.1f,0.7f);

glEnd();


glBegin(GL_QUADS);

glColor3f(1.0f, 1.0f, 1.0f);

glVertex2f(0.1f,0.1f);

glVertex2f(0.1f,0.3f);

glVertex2f(0.3f,0.3f);

glVertex2f(0.3f,0.1f);

glEnd();

glBegin(GL_QUADS);

glColor3f(1.0f, 1.0f, 1.0f);

glVertex2f(0.5f,0.1f);

glVertex2f(0.5f,0.3f);

glVertex2f(0.9f,0.3f);
```

```
glVertex2f(0.9f,0.1f);
glEnd();
glBegin(GL_QUADS);
glColor3f(1.0f, 1.0f, 1.0f);
glVertex2f(0.9f,0.5f);
glVertex2f(0.9f,0.7f);
glVertex2f(0.5f,0.7f);
glVertex2f(0.5f,0.5f);
glEnd();

glBegin(GL_QUADS);
glColor3f(1.0f, 1.0f, 1.0f);
glVertex2f(0.3f,0.7f);
glVertex2f(0.3f,0.5f);
glVertex2f(0.1f,0.5f);
glVertex2f(0.1f,0.7f);
glEnd();
```

//Georgia

```
glBegin(GL_QUADS);
glColor3f(1.0f, 0.0f, 0.0f);
glVertex2f(0.1f,-0.1f);
glVertex2f(0.1f,-0.7f);
glVertex2f(0.9f,-0.7f);
glVertex2f(0.9f,-0.1f);
glEnd();

glBegin(GL_QUADS);
glColor3f(1.0f, 1.0f, 1.0f);
glVertex2f(0.1f,-0.1f);
glVertex2f(0.4f,-0.1f);
glVertex2f(0.4f,-0.3f);
glVertex2f(0.1f,-0.3f);
glEnd();

glLineWidth(3.5f);
glBegin(GL_LINES);
glColor3f(1.0f, 0.0f, 0.0f);
glVertex2f(0.25f,-0.16f);
glVertex2f(0.25f,-0.24f);
glEnd();
glLineWidth(3.5f);
glBegin(GL_LINES);
glColor3f(1.0f, 0.0f, 0.0f);
glVertex2f(0.22f,-0.2f);
glVertex2f(0.28f,-0.2f);
```

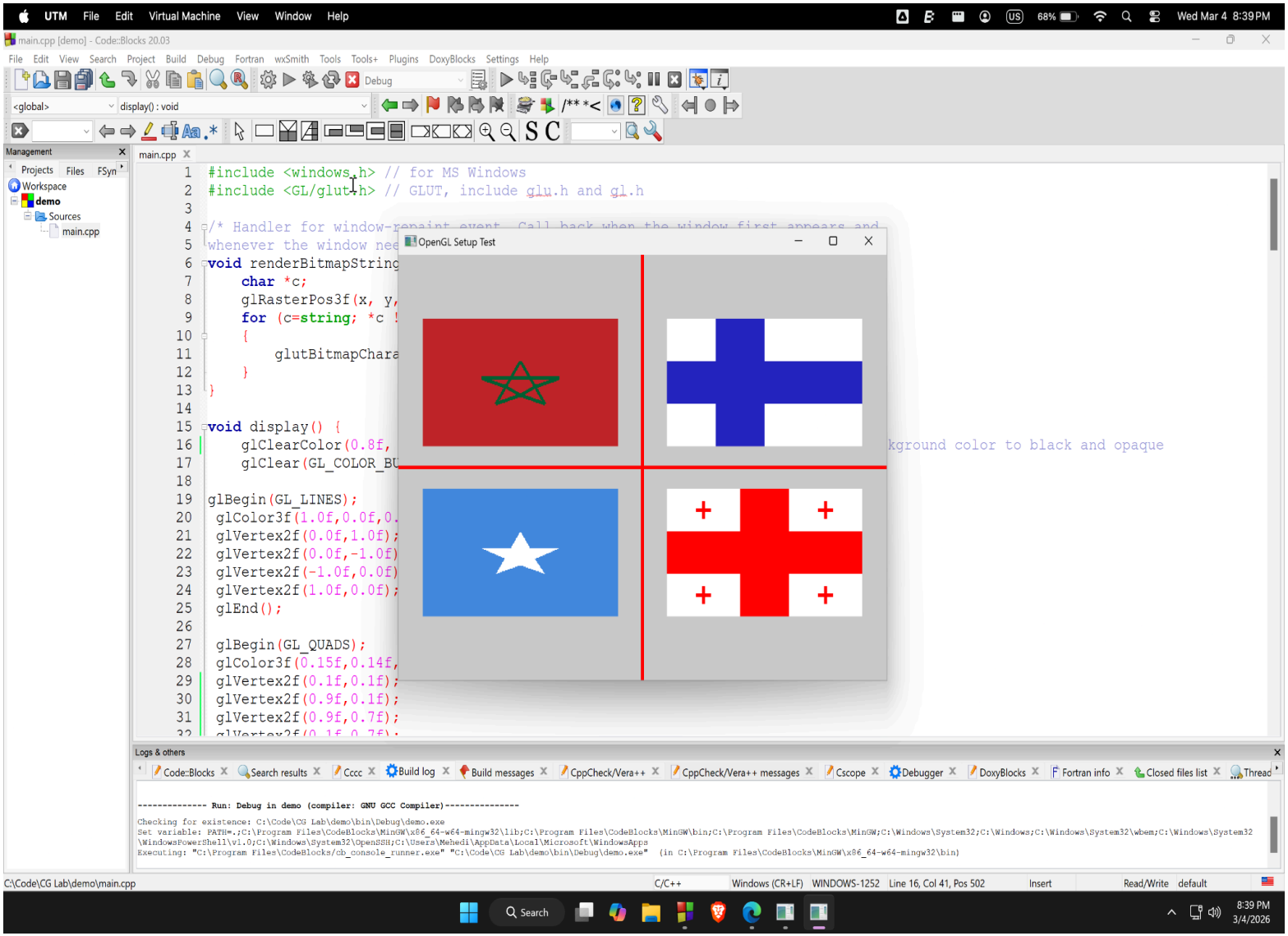
```
glEnd();

 glBegin(GL_QUADS);
glColor3f(1.0f, 1.0f, 1.0f);
glVertex2f(0.6f,-0.1f);
glVertex2f(0.9f,-0.1f);
glVertex2f(0.9f,-0.3f);
glVertex2f(0.6f,-0.3f);
glEnd();

glLineWidth(3.5f);
glBegin(GL_LINES);
glColor3f(1.0f, 0.0f, 0.0f);
glVertex2f(0.75f,-0.16f);
glVertex2f(0.75f,-0.24f);
glEnd();

glLineWidth(3.5f);
glBegin(GL_LINES);
glColor3f(1.0f, 0.0f, 0.0f);
glVertex2f(0.72f,-0.2f);
glVertex2f(0.78f,-0.2f);
glEnd();

glBegin(GL_QUADS);
glColor3f(1.0f, 1.0f, 1.0f);
glVertex2f(0.1f,-0.5f);
glVertex2f(0.4f,-0.5f);
glVertex2f(0.4f,-0.7f);
glVertex2f(0.1f,-0.7f);
glEnd();

glLineWidth(3.5f);
glBegin(GL_LINES);
glColor3f(1.0f, 0.0f, 0.0f);
glVertex2f(0.25f,-0.56f);
glVertex2f(0.25f,-0.64f);
glEnd();

glLineWidth(3.5f);
glBegin(GL_LINES);
glColor3f(1.0f, 0.0f, 0.0f);
glVertex2f(0.22f,-0.6f);
glVertex2f(0.28f,-0.6f);
glEnd();
```

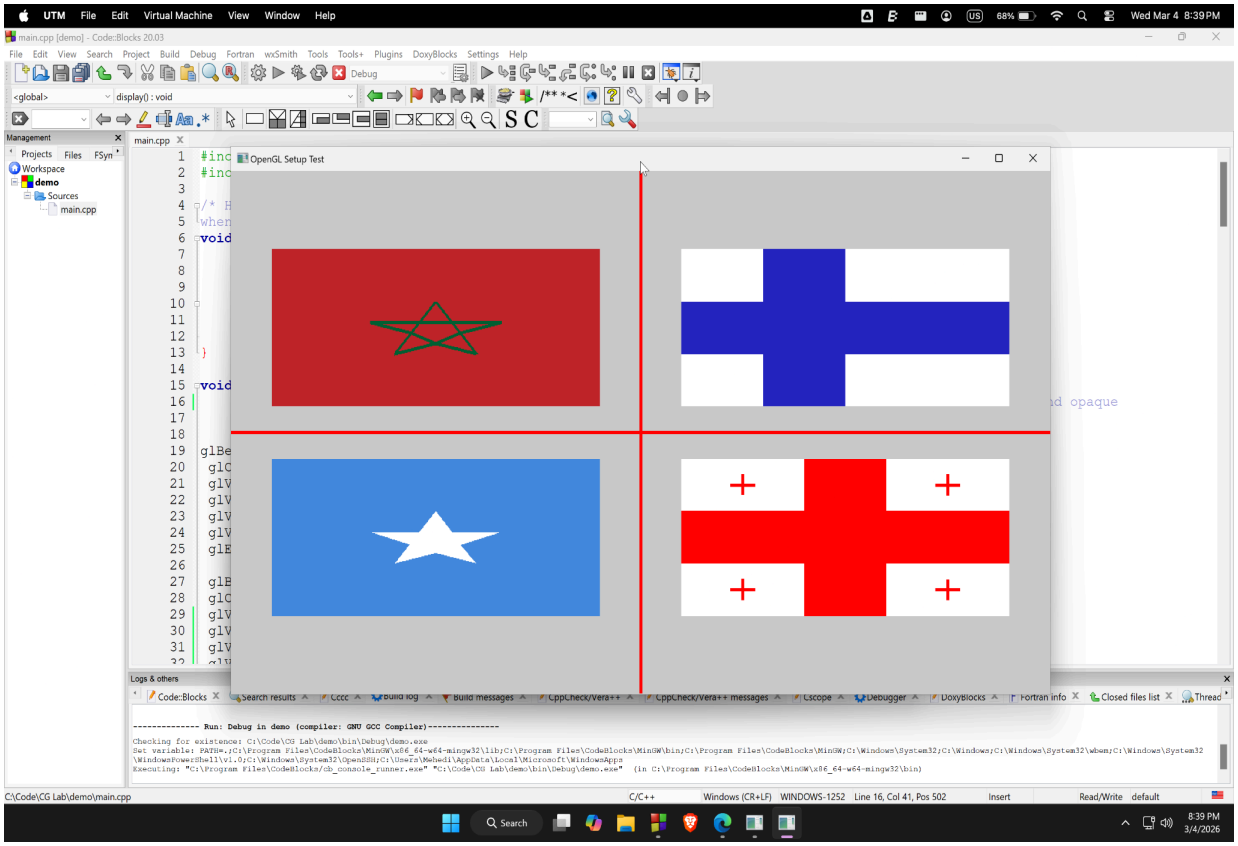
```
 glBegin(GL_QUADS);
glColor3f(1.0f, 1.0f, 1.0f);
glVertex2f(0.6f,-0.5f);
glVertex2f(0.9f,-0.5f);
glVertex2f(0.9f,-0.7f);
glVertex2f(0.6f,-0.7f);
glEnd();

glLineWidth(3.5f);
glBegin(GL_LINES);
glColor3f(1.0f, 0.0f, 0.0f);
glVertex2f(0.75f,-0.56f);
glVertex2f(0.75f,-0.64f);
glEnd();

glLineWidth(3.5f);
glBegin(GL_LINES);
glColor3f(1.0f, 0.0f, 0.0f);
glVertex2f(0.72f,-0.6f);
glVertex2f(0.78f,-0.6f);
glEnd();
```

//Morocco

```
glBegin(GL_QUADS);
glColor3f(0.757f, 0.153f, 0.176f);
glVertex2f(-0.1f,0.1f);
glVertex2f(-0.9f,0.1f);
glVertex2f(-0.9f,0.7f);
glVertex2f(-0.1f,0.7f);
glEnd();

glLineWidth(3.5f);
glBegin(GL_LINE_LOOP);
glColor3f(0.0f, 0.384f, 0.2f);
glVertex2f(-0.5f,0.5f);
glVertex2f(-0.54f,0.42f);
glVertex2f(-0.66f,0.42f);
glVertex2f(-0.56f,0.38f);
glVertex2f(-0.6f,0.3f);
glVertex2f(-0.5f,0.34f);
glVertex2f(-0.4f,0.3f);
glVertex2f(-0.44f,0.38f);
glVertex2f(-0.34f,0.42f);
glVertex2f(-0.46f,0.42f);
glVertex2f(-0.5f,0.5f);
```

```
glVertex2f(-0.54f,0.42f);
glVertex2f(-0.56f,0.38f);
glVertex2f(-0.5f,0.34f);
glVertex2f(-0.44f,0.38f);
glVertex2f(-0.46f,0.42f);
glVertex2f(-0.54f,0.42f);
glEnd();

//Somalia
glBegin(GL_QUADS);
glColor3f(0.255f, 0.537f, 0.867f);
glVertex2f(-0.1f,-0.1f);
glVertex2f(-0.9f,-0.1f);
glVertex2f(-0.9f,-0.7f);
glVertex2f(-0.1f,-0.7f);
glEnd();

glBegin(GL_POLYGON);
glColor3f(1.0f, 1.0f, 1.0f);
//glVertex2f(-0.5f,-0.3f);
glVertex2f(-0.54f,-0.38f);
glVertex2f(-0.66f,-0.38f);
glVertex2f(-0.56f,-0.43f);
glVertex2f(-0.6f,-0.5f);
glVertex2f(-0.5f,-0.46f);
glVertex2f(-0.4f,-0.5f);
glVertex2f(-0.44f,-0.43f);
glVertex2f(-0.34f,-0.38f);
glVertex2f(-0.46f,-0.38f);
glVertex2f(-0.5f,-0.3f);
glEnd();
glFlush(); // Render now
}
int main(int argc, char** argv)
{
   glutInit(&argc, argv);
   glutInitWindowSize(640, 480);
glutInitWindowPosition(80, 50);
   glutCreateWindow("OpenGL Setup
Test");
glutDisplayFunc(display);
   glutMainLoop();
return 0; }
```


UTM File Edit Virtual Machine View Window Help
Wed Mar 4 8:39 PM
main.cpp [demo] - Code::Blocks 20.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
Debug
<global>
display() : void
Management
Projects Files FSyn
Workspace
demo
Sources
main.cpp
main.cpp
#include <windows.h> // for MS Windows
#include <GL/glut.h> // GLUT, include glu.h and gl.h
/* Handler for window-repaint event. Call back when the window first appears and
whenever the window nee
void renderBitmapString
char *c;
glRasterPos3f(x, y,
for (c=string; *c !
{
glutBitmapChara
}
}
void display() {
glClearColor(0.8f,
glClear(GL_COLOR_BU
kground color to black and opaque
glBegin(GL_LINES);
glColor3f(1.0f,0.0f,0.
glVertex2f(0.0f,1.0f);
glVertex2f(0.0f,-1.0f)
glVertex2f(-1.0f,0.0f)
glVertex2f(1.0f,0.0f);
glEnd();
glBegin(GL_QUADS);
glColor3f(0.15f,0.14f,
glVertex2f(0.1f,0.1f);
glVertex2f(0.9f,0.1f);
glVertex2f(0.9f,0.7f);
OpenGL Setup Test
Logs & others
Code::Blocks Search results Cccc Build log Build messages CppCheck/Vera++ CppCheck/Vera++ messages Cscope Debugger DoxyBlocks Fortran info Closed files list Thread
---------- Run: Debug in demo (compiler: GNU GCC Compiler)----------
Checking for existence: C:\Code\CG Lab\demo\bin\Debug\demo.exe
Set variable: PATH=.;C:\Program Files\CodeBlocks\MinGW\x86_64-w64-mingw32\lib;C:\Program Files\CodeBlocks\MinGW\bin;C:\Program Files\CodeBlocks\MinGW;C:\Windows\System32;C:\Windows;C:\Windows\System32\wbem;C:\Windows\System32
\WindowsPowerShell\v1.0;C:\Windows\System32\OpenSSH;C:\Users\Mehedi\AppData\Local\Microsoft\WindowsApps
Executing: "C:\Program Files\CodeBlocks/cb_console_runner.exe" "C:\Code\CG Lab\demo\bin\Debug\demo.exe" (in C:\Program Files\CodeBlocks\MinGW\x86_64-w64-mingw32\bin)
C:\Code\CG Lab\demo\main.cpp
C/C++ Windows (CR+LF) WINDOWS-1252 Line 16, Col 41, Pos 502 Insert Read/Write default
Search
8:39 PM
3/4/2026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File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
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<global>
display() : void
Management
Projects Files FSym
Workspace
demo
Sources
main.cpp
OpenGL Setup Test
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++
CppCheck/Vera++ messages
Cscope
Debugger
DoxyBlocks
Fortran info
Closed files list
Thread
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Set variable: PATH=.;C:\Program Files\CodeBlocks\MinGW\x86_64-w64-mingw32\lib;C:\Program Files\CodeBlocks\MinGW\bin;C:\Program Files\CodeBlocks\MinGW;C:\Windows\System32;C:\Windows;C:\Windows\System32\wbem;C:\Windows\System32\WindowsPowerShell\v1.0;C:\Windows\System32\OpenSSH;C:\Users\Mehedi\AppData\Local\Microsoft\WindowsApps
Executing: "C:\Program Files\CodeBlocks/cb_console_runner.exe" "C:\Code\CG Lab\demo\bin\Debug\demo.exe" (in C:\Program Files\CodeBlocks\MinGW\x86_64-w64-mingw32\bin)
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C/C++
Windows (CR+LF)
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Line 16, Col 41, Pos 502
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